

Parth Mital

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🐙 github.com/parthmital 🌐 parthmital-portfolio.vercel.app/

Education

Vellore Institute of Technology (VIT)

B.Tech in Computer Science & Engineering (CGPA: 8.75 / 10)

2023 – 2027

Vellore, India

Projects

🔗 NeuroScan AI: Brain Tumour MRI Analysis System

- Conducted comparative research on **21 deep learning architectures** for brain tumour analysis.
- Authored **2 research papers (under review)** on performance, convergence & efficiency analysis.
- Achieved **99.59% detection accuracy, 98.58% classification accuracy & 0.79 Dice score** for **3D segmentation**.
- Engineered **end-to-end AI platform** for automated MRI inference, 3D visualisation & report generation.
- **Tech Stack:** Python, PyTorch, FastAPI, SQLAlchemy, SQLite, OpenCV, NiBabel, React, TypeScript, Vite, Tailwind CSS, VTK.js, JWT.

🔗 PPE Compliance Monitor: Real-Time Safety Detection System

- Developed **real-time PPE compliance monitoring system** using **deep learning based object detection**.
- Achieved **0.88 mAP@0.5, 0.64 mAP@0.5:0.95, 92.27% precision & 80.92% recall** on test data.
- Optimised inference to **23 ms per image (40+ FPS)** with **temporal buffering** for reduced false positives.
- Delivered **end-to-end system** with image & video processing, async pipelines, incident logging & analytics dashboard.
- **Tech Stack:** Python, PyTorch, YOLOv8, FastAPI, OpenCV, SQLite, React, TypeScript, Vite, Tailwind CSS.

🔗 The Obsidian Flask: LLM NPC Dialogue Engine

- Engineered **RAG-based dialogue engine** enabling **context-aware NPC interactions** with persistent memory.
- Designed **7-stage deterministic pipeline** for **retrieval, validation & structured LLM outputs**.
- Scaled memory with **dual-tier system** (8-turn context + **top-6 FAISS retrieval over 1000+ interactions**).
- Implemented **event-sourced state management** with **full audit trails & replayable game state**.
- **Tech Stack:** Python, FastAPI, LangGraph, FAISS, Sentence Transformers, SQLite, WebSockets, React, TypeScript, Zustand, Vite, Tailwind CSS, Docker.

🔗 PDF Renamer: PDF Title Generation System

- Developed **T5-based sequence-to-sequence model** for **automated PDF title generation** from document text.
- Curated **custom dataset of 315 PDFs** with **manual ground truth annotations**.
- Achieved **0.275 BLEU, 0.446 ROUGE-1 & 20.2% exact match** on test evaluation.
- **Tech Stack:** Python, Hugging Face Transformers, T5, PyTorch, pdfplumber, Datasets, NumPy, Pandas.

Technical Skills

Languages: C, C++, Java, Python, TypeScript, JavaScript

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Digital Systems Design, Microprocessors & Microcontrollers

Software Development: Full-Stack Development, Frontend, Backend, Databases, REST APIs, Authentication

AI/ML & Data Science: Machine Learning, Deep Learning, Model Training & Evaluation, Computer Vision, NLP, Feature Engineering, EDA, Data Processing, ETL Pipelines

Cloud & DevOps: Containerization, Cloud Deployment, AWS, Docker

Experience

Hiraeth Tech

Web Development Intern

Jun 2025 – Jul 2025

Vadodara, India

- Built **5+ responsive WordPress websites** using Elementor & custom CSS.
- Optimised images & assets to improve page load performance.
- Ensured cross-device responsiveness & consistent UI/UX.